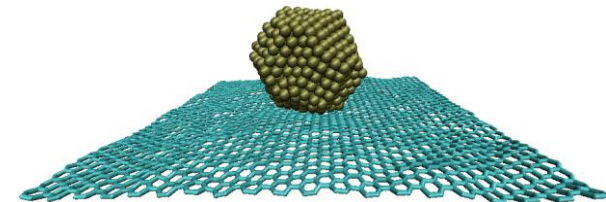


Topic 3: Zero-Dimensional Nanostructures

Questions we will answer...

- What is the difference between top-down and bottom-up methods of materials synthesis?
- What are the criteria for quantum confinement?
- How are nanoparticles synthesized?
- What are desirable nanoparticle characteristics?
- How does homogeneous nucleation occur?
- How are nucleation and growth connected to processing conditions?
- How is nanoparticle size determined?
- What is heterogeneous nucleation?



Source: Schebarchov

Zero-Dimensional Nanostructures are...

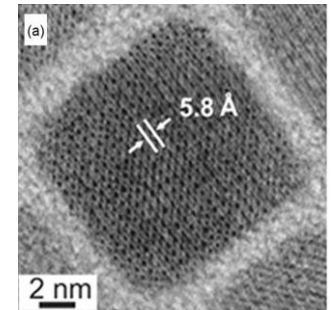
Nanoparticles ranging in size from ~ 1 nm to ~ 200 nm and in shape from spheres, cubes, trapezoids

Single crystals (*aka* nanocrystals), polycrystalline, amorphous, and may have core-shell structures

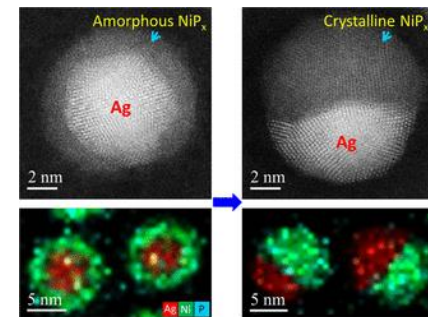
Metals, semiconductors, oxides, carbon-based materials

Quantum dots, if quantum confinement is felt

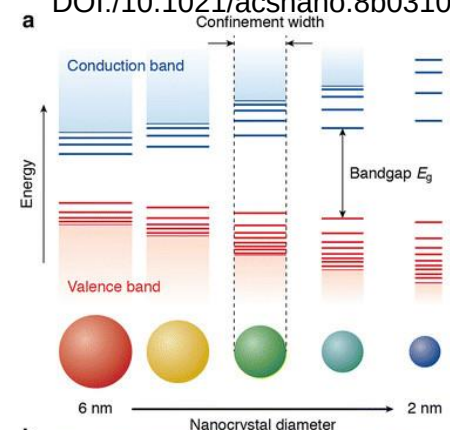
Made from *bottom-up* approaches, such as solution- and vapor-phase synthesis, or from *top-down* approaches, including attrition, milling, lithography



DOI:10.1016/j.scib.2017.01.006



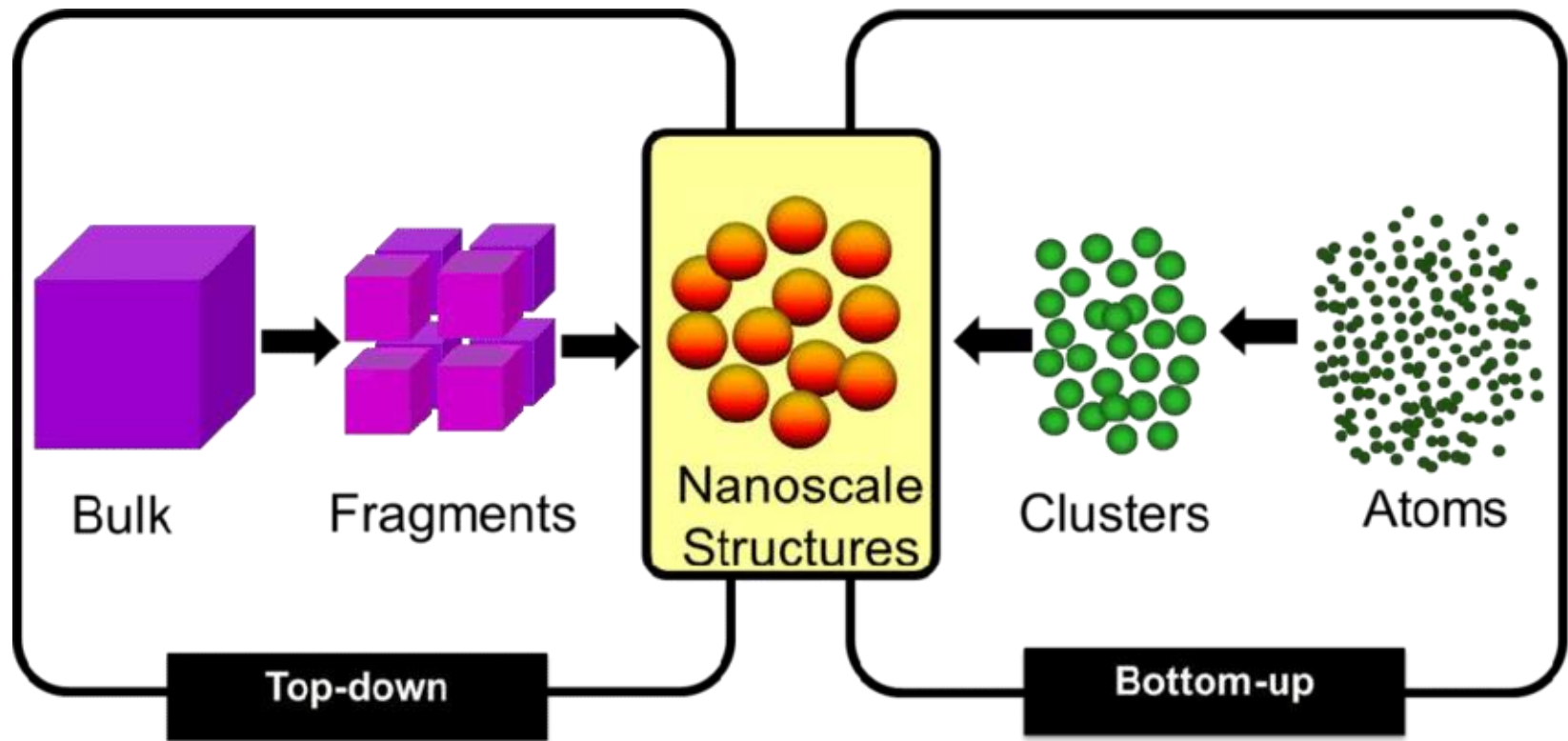
DOI:10.1021/acsnano.8b03106



DOI:10.1021/acs.jpcclett.7b01640

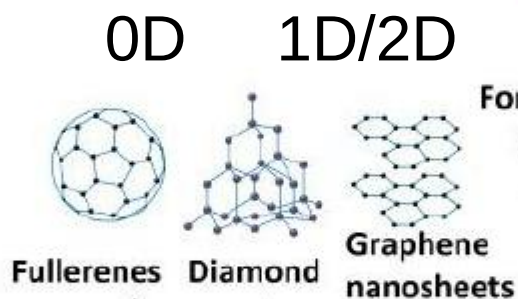
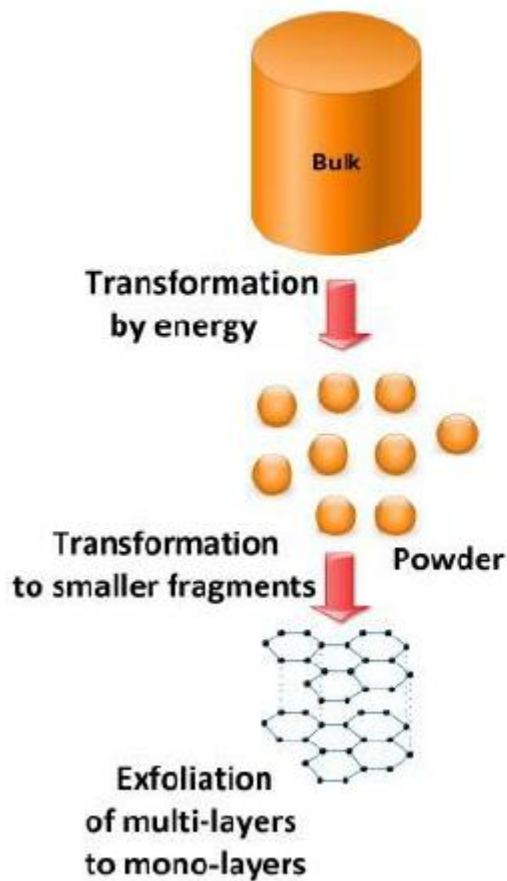
Bottom-up vs Top-down

Top-down methods of nanomaterial synthesis



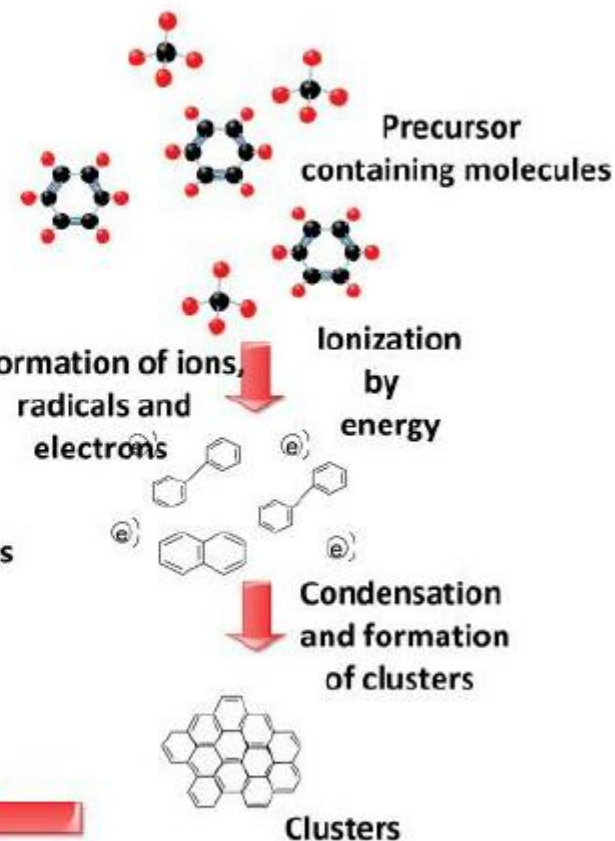
Bottom-up vs Top-down

Top-Down Approach

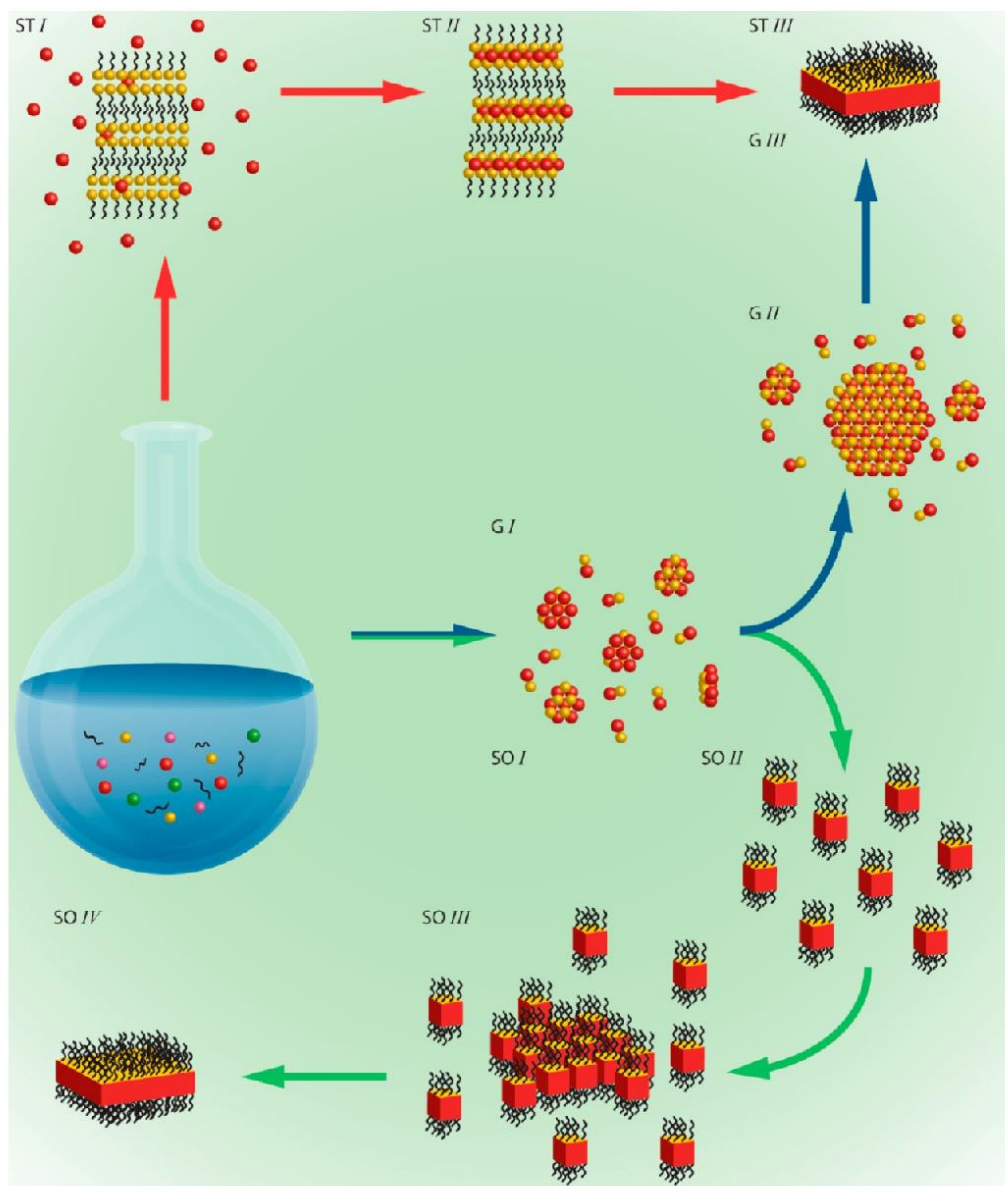


Nanoparticles Synthesized

Bottom-Up Approach



Material Formation Pathways (with a Ligand)



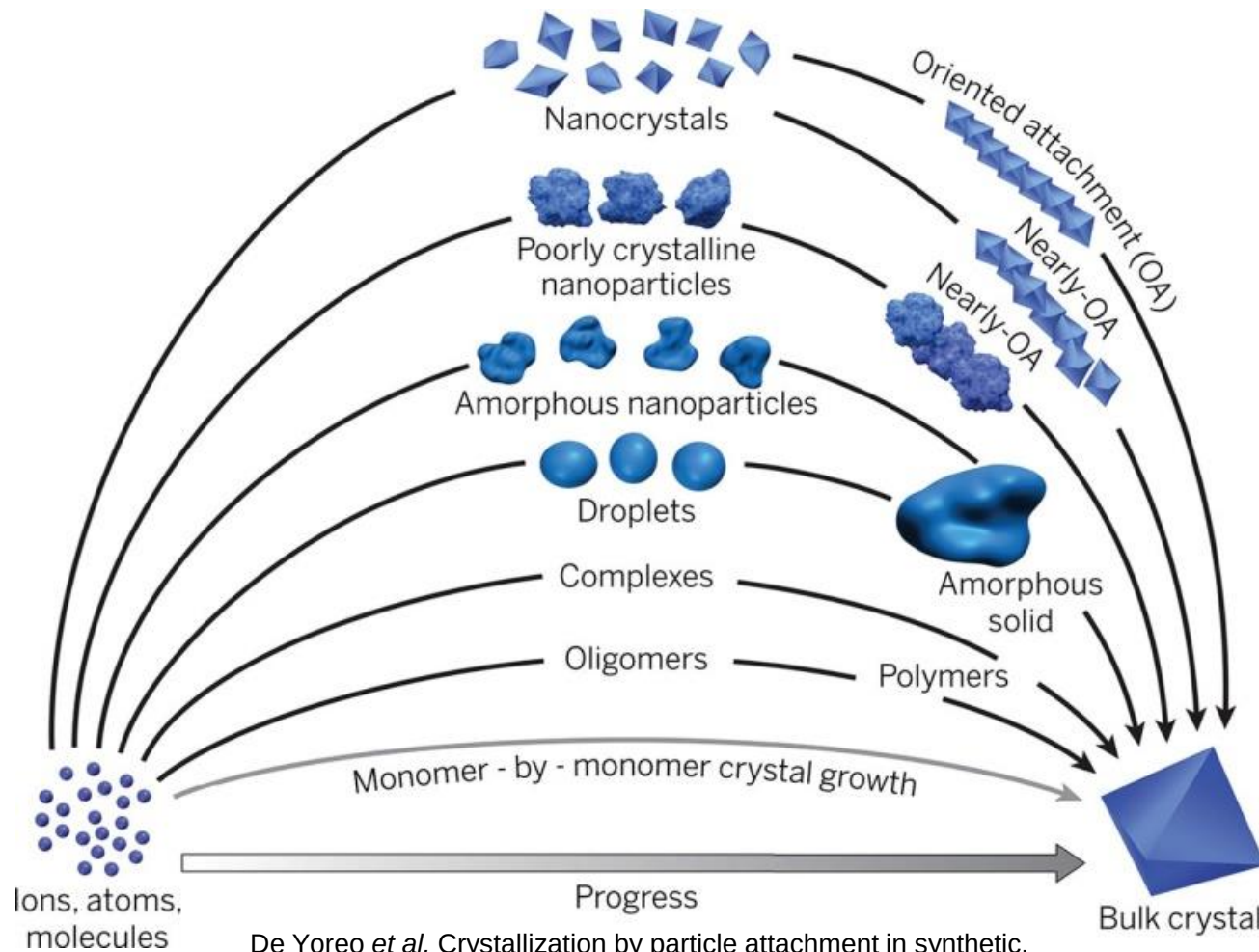
“Schematic overview of possible formation mechanisms for ultrathin 2D NCs.

The first pathway, marked with red arrows, shows the subsequent steps in the soft template (ST) mechanism, in which ligand chains order (ST I) and either constrain nucleation and growth in 2D, or direct the oriented attachment of building blocks in the lateral directions (ST II).

The second pathway is marked with blue arrows and shows 2D-constrained growth (G) of magic size cluster seeds (G I) that grow only in lateral dimensions (G II) due to the directive effect of ligands (not shown for clarity) and/or facet reactivity.

The third pathway, indicated by green arrows, shows NS formation through self-organization (SO). Nucleation (SO I) and growth of NC building blocks (SO II) is followed by self-organization (SO III) and oriented attachment (SO IV), directed by dense ligand layers on certain facets. After self-organizing into a 2D superstructure, the NCs fuse into a single-crystalline NS.”

Material Formation Pathways



De Yoreo *et al.* Crystallization by particle attachment in synthetic, biogenic, and geologic environments. *Science* 349 (2015), aaa6760.

Aggregative

Nonclassical nucleation

Classical nucleation

Our focus

Bottom-Up Nanoparticle Preparation

Synthesis and preparation methods can be grouped in two main categories:

- Thermodynamic equilibrium approach
- Kinetic approach

Thermodynamic synthesis approach:

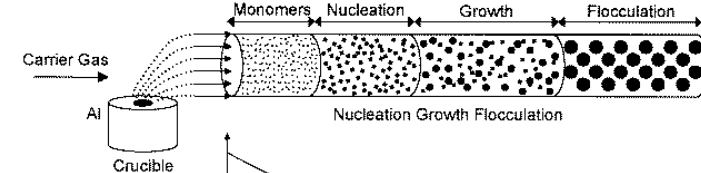
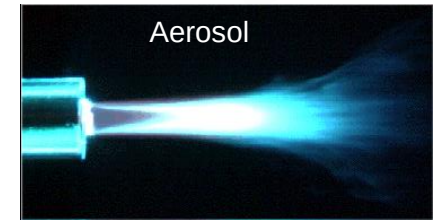
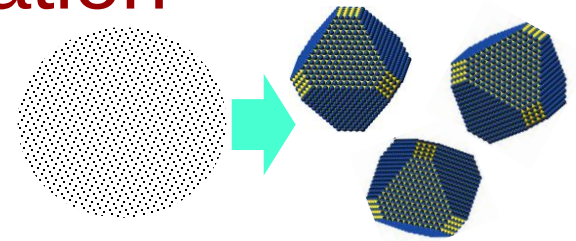
1. Generation of supersaturation
2. Nucleation
3. Growth

Our focus

Kinetic synthesis approach:

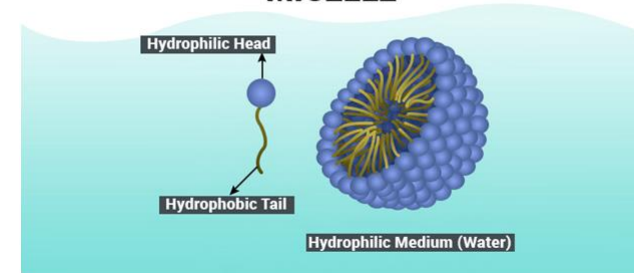
- Limited amount of precursors available for growth
- Spatial confinement of synthesis, e.g., aerosol, micelle

Solution-based approach will be emphasized, but fundamentals and principles are applicable to other systems (e.g., vapor, solid)



http://www.umich.edu/~elements/w eb_mod/aerosol/

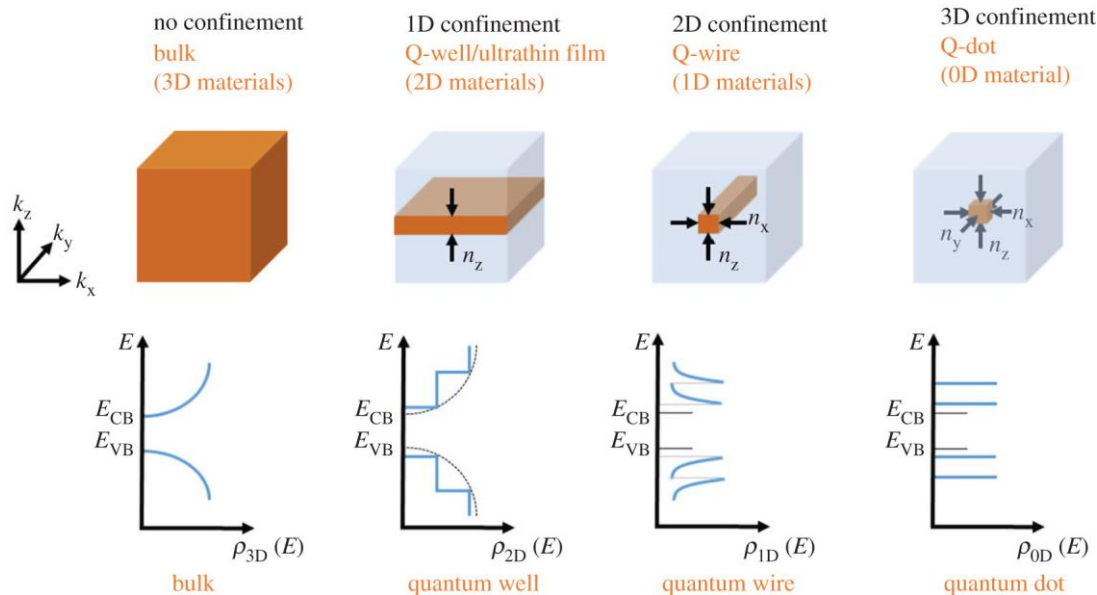
MICELLE



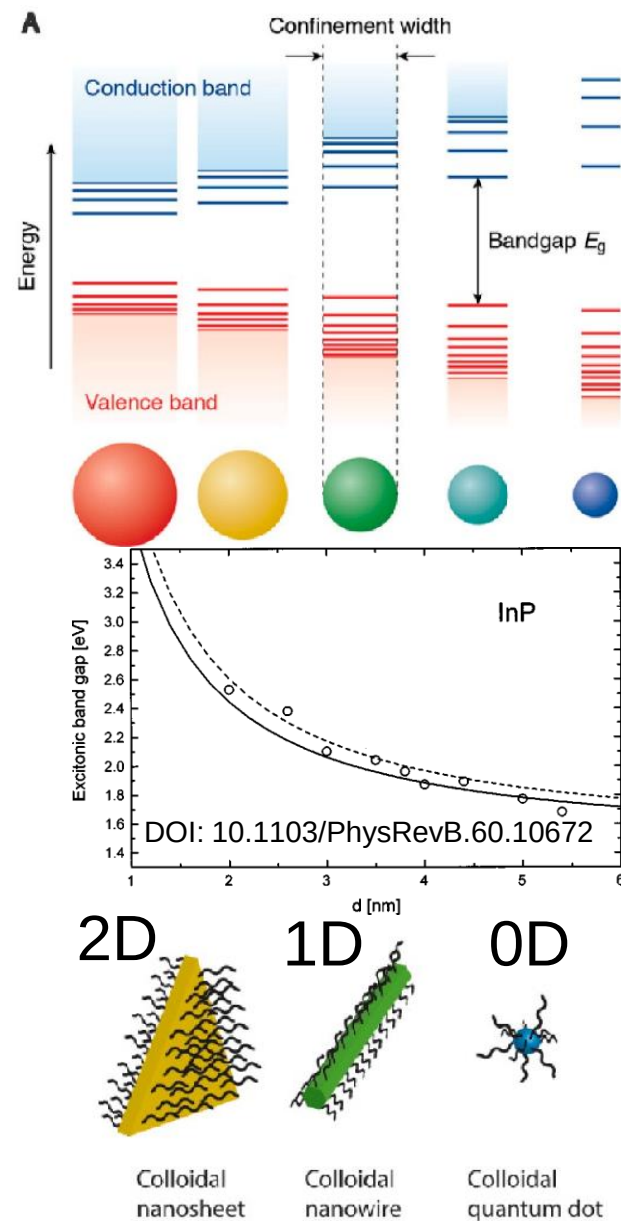
Quantum Confinement

When the size of a nanocrystal approaches the *de Broglie* wavelength of electrons (wave-particle duality) in the conduction band, e^- and h^+ are spatially confined, dipoles are formed and discrete electronic energy levels arise, and shift E_g (band gap). This is known as *quantum confinement*

This is most pronounced for semiconductor NPs (2-10 nm), but also observed in metals (<2 nm) and insulators



<https://doi.org/10.1098/rsos.180387>



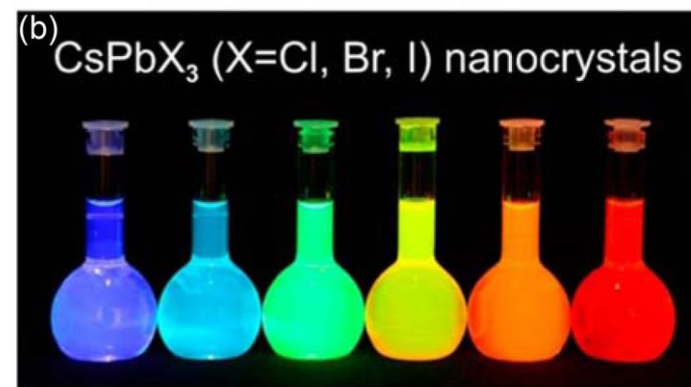
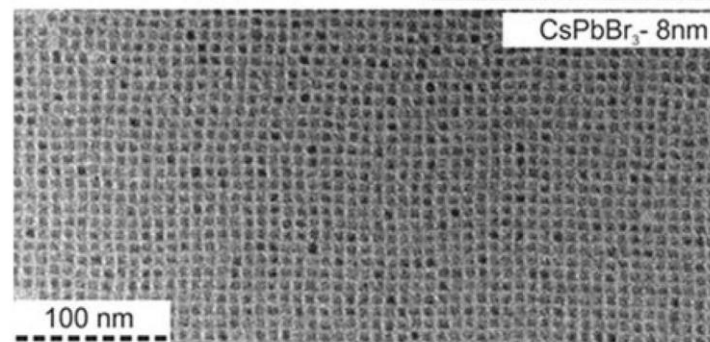
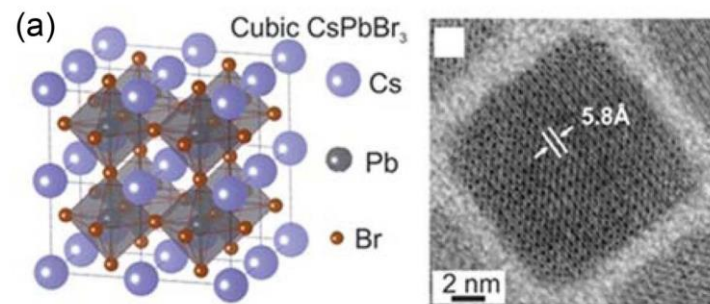
DOI:10.1021/acs.jpcclett.7b01640

Desirable Nanoparticle Characteristics

Small size is not the only requirement. For any practical application, *synthesis & stability* needs to be controlled in such a way that the resulting NPs have the following characteristics:

Identical and stable properties, which requires:

1. Identical size of all particles, *aka* monosized or uniform size distribution
2. Identical shape or morphology
3. Identical chemical composition
4. Identical crystal structure
5. Core and surface compositions must be the same
6. Individually dispersed or monodispersed, i.e., no agglomeration
7. If agglomeration occurs, particles must be readily redispersible



DOI:10.4191/kcers.2018.55.6.11

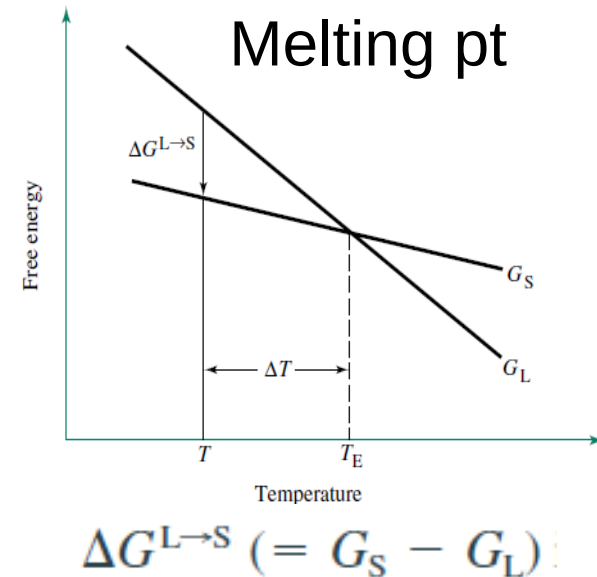
Homogeneous Nucleation

NPs can be synthesized through *homogeneous nucleation* in three media: liquid, gas and solid.

The fundamentals of nucleation and subsequent growth are the same, so *we will focus on solutions*.

Nucleation requires supersaturation of growth species, *i.e.*, the concentration of solute to exceed the equilibrium solubility (or temperature to decrease below the phase transformation point), for example:

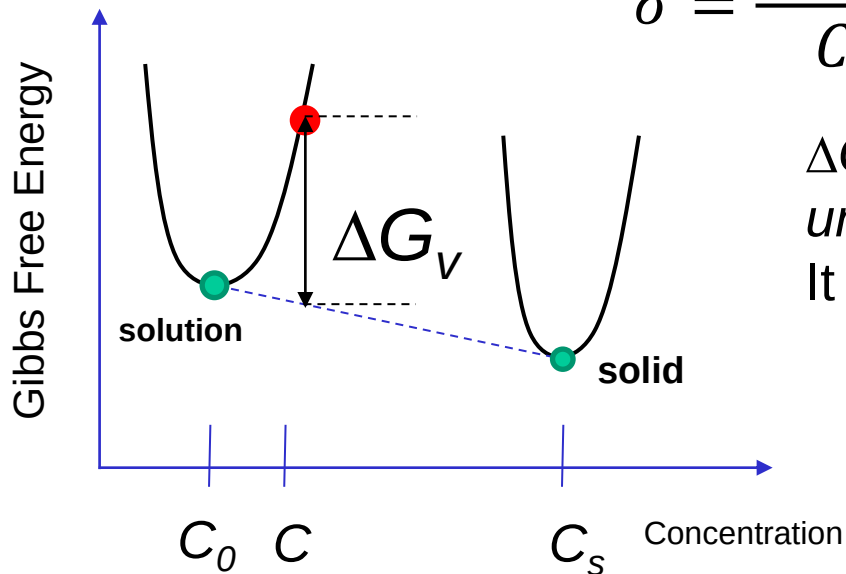
- Reduction of temperature in most systems reduces solubility and can cause supersaturation;
Note: some materials exhibit an inverse temperature solubility behavior
- *In situ* chemical reactions can transform precursors and change their solubility, leading to supersaturation
- Solvent can evaporate, leading to supersaturation, but this approach is not used for controlled NP nucleation and growth



Homogeneous Nucleation

Let's consider a solution with a solute concentration, C , and an equilibrium concentration, C_0 . The supersaturation of the solution, σ , is defined as

$$\sigma = \frac{C - C_0}{C_0}$$



ΔG_v is the change of Gibbs free energy *per unit volume* of formation of the solid phase
It depends on the concentration of solute:

$$\Delta G_v = \frac{-kT}{\Omega} \ln \sigma$$

Ω : atomic volume

Without supersaturation, $\sigma = 0 \rightarrow \Delta G = 0 \rightarrow$ No nucleation!

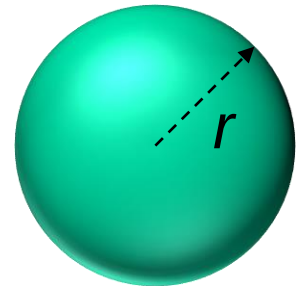
When $C > C_0$, $\sigma > 0$, $\rightarrow \Delta G < 0 \rightarrow$ nucleation wants to occur.

Homogeneous Nucleation

Assuming a spherical nucleus with a radius r is formed, the change in Gibbs free energy associated to the volume can be described as

$$\Delta G_{volume} = \frac{4}{3} \pi r^3 \Delta G_v$$

The energy reduction due to the solid volume of the NP is *counterbalanced by the introduction of a surface*, and its associated surface energy per unit area, γ .



$$V = \frac{4}{3} \pi r^3$$

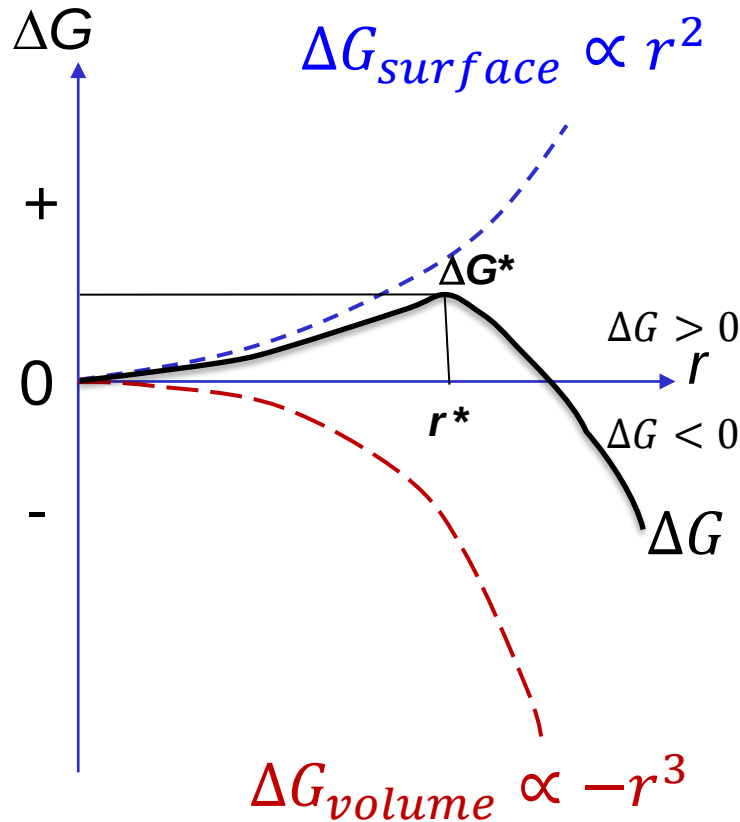
$$S = 4\pi r^2$$

The total change of Gibbs free energy, ΔG , is given by

$$\Delta G = \Delta G_{volume} + \Delta G_{surface} = \frac{4}{3} \pi r^3 \Delta G_v + 4\pi r^2 \gamma$$

Note: $\Delta G_v < 0$, while $\gamma > 0$

Homogeneous Nucleation



The newly formed nucleus is stable only when $r \geq r^*$, which is known as the critical radius

A nucleus smaller than r^* will dissolve into the solution to reduce ΔG , but a nucleus larger than r^* will grow to reduce ΔG .

Nucleation therefore exhibits an activation barrier, ΔG^* , which must be overcome for homogeneous nucleation to occur. This is known as the critical energy.

Note that ΔG peaks when $r = r^*$, therefore $\frac{d(\Delta G)}{dr} = 0$

$$\frac{d(\Delta G)}{dr} = 4\pi r^2 \Delta G_v + 8\pi r \gamma = 0, r = r^*$$

$$\Delta G^* = \frac{4}{3} \pi r^{*3} \Delta G_v + 4\pi r^{*2} \gamma$$

$$r^* = -2 \frac{\gamma}{\Delta G_v}$$

$$\Delta G^* = \frac{16\pi}{3} \frac{\gamma^3}{\Delta G_v^2}$$

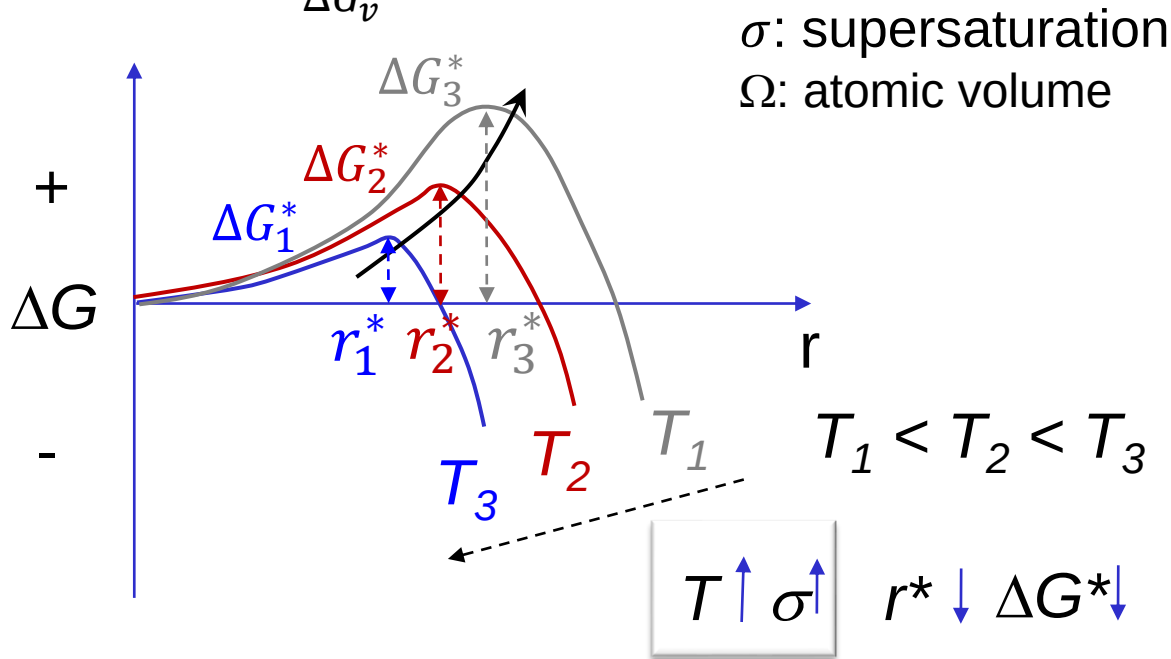
$$\Delta G^* = \frac{4\pi}{3} \gamma r^{*2}$$

Homogeneous Nucleation

In synthesizing NPs, r^* represents the limit of how small NPs can be synthesized.

To reduce r^* and ΔG^* , we need to increase the magnitude of the negative term, i.e., ΔG_v :

$$\Delta G^* = \frac{16\pi}{3} \frac{\gamma^3}{\Delta G_v^2} \quad \Delta G_v = \frac{-kT}{\Omega} \ln \sigma \quad \Delta G^* = \frac{16\pi\Omega^2}{3(kT)^2} \frac{\gamma^3}{(\ln \sigma)^2} \quad r^* = \frac{2\Omega}{kT} \frac{\gamma}{\ln \sigma}$$

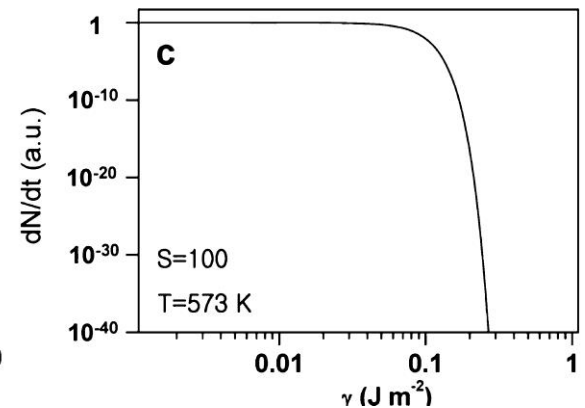
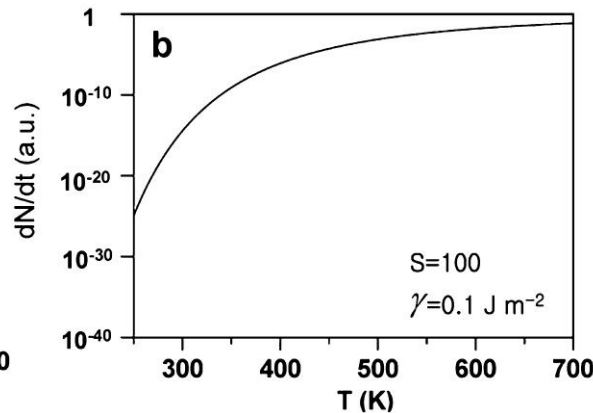
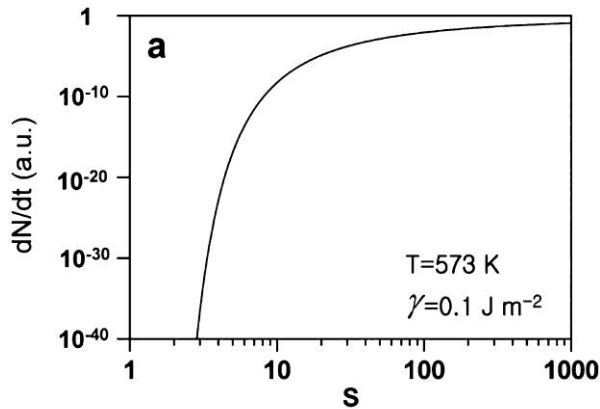


Note: Surface energy can vary with T , especially near roughening temperature of a NC, but its effect is ignored.

Homogeneous Nucleation

The rate of nucleation, R_N , of N particles per unit time t is described by an Arrhenius type equation, with pre-exponential factor A , *which we will ignore here*:

$$R_N = \frac{dN}{dt} = A e^{-\frac{\Delta G^*}{kT}} = A e^{-\frac{16\pi\Omega^2 \gamma^3}{3(kT)^3 (\ln \sigma)^2}}$$



$\sigma : S$

CdSe; assume $A = 1$

$\gamma = 3.29 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}$

The largest effect on nucleation rate comes from **supersaturation**, where a change from $\sigma = 2$ to $\sigma = 4$ causes an increase in the nucleation rate about $\sim 10^{70}$, and next by a variation in the **surface free energy** caused by different surfactants, followed by **reaction temperature**.

Homogeneous Nucleation

R_N is also proportional to the number of growth species per unit volume, n , which can be considered as nucleation centers. n is the initial concentration, C_0 , of the solution for homogeneous nucleation.

Finally, R_N is proportional to the jump frequency, Γ , of growth species from one site to the other.

$$\Gamma = \frac{kT}{3\pi\lambda^3\eta}$$

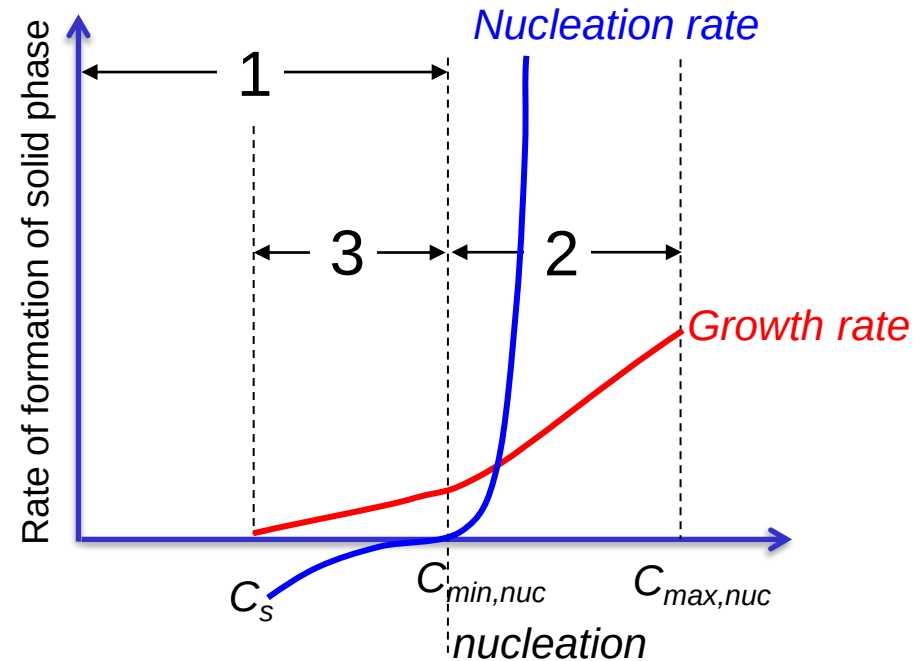
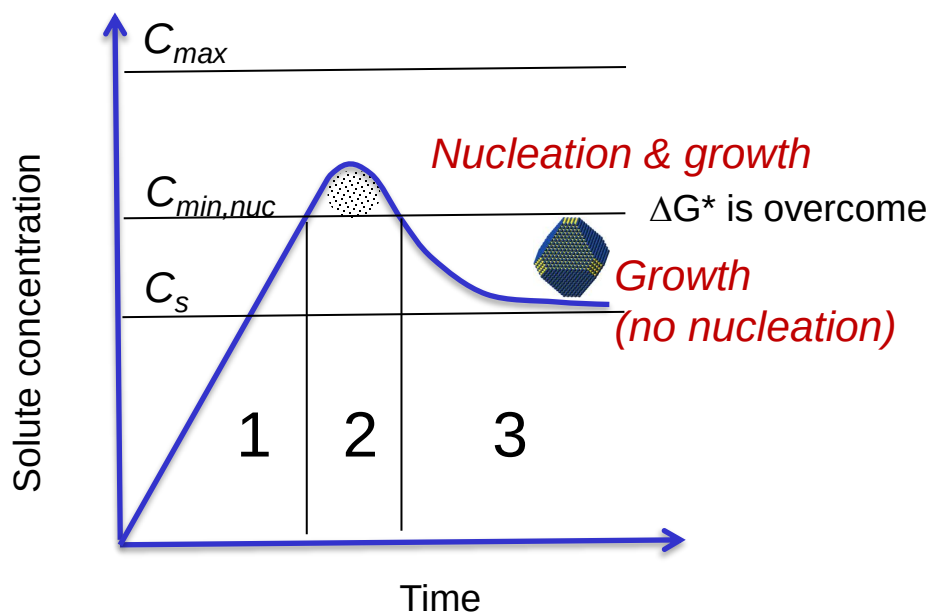
λ : diameter of growth species
 η : viscosity of the solution

$n\Gamma = A$ constitutes the pre-exponential factor.

$$R_N = \frac{dN}{dt} = n\Gamma e^{-\frac{\Delta G^*}{kT}} = \frac{C_0 kT}{3\pi\lambda^3\eta} e^{-\frac{16\pi\Omega^2 \gamma^3}{3(kT)^3 (\ln \sigma)^2}}$$

A **high initial concentration**, C_0 , a high supersaturation, σ , and **low viscosity**, η , favor the formation of a **larger number of nuclei**.

Growth Mechanism of Nuclei



Nucleation and growth occur at different rates and in different concentration regimes, e.g. supersaturation.

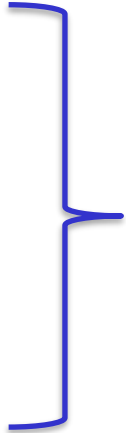
For synthesis of NPs with uniform size distribution, it is best if all nuclei form at the same time and with the same size, and therefore over a *very short time period*. This provides the highest likelihood that nuclei will have the same size and the same subsequent growth.

Size distribution is further altered during *subsequent growth* and requires process control.

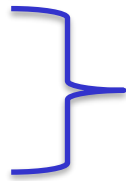
Growth Mechanism of Multiple Nuclei

Major steps of nanoparticle growth:

- Generation of growth species
- Diffusion of growth species to the growth surface
- Adsorption (reversible) of growth species into growth surface
- Incorporation (irreversible) of adsorbates into growth surface



If supply to the surface is limited:
diffusion-limited



If abundant supply but incorporation limited:
Growth-limited

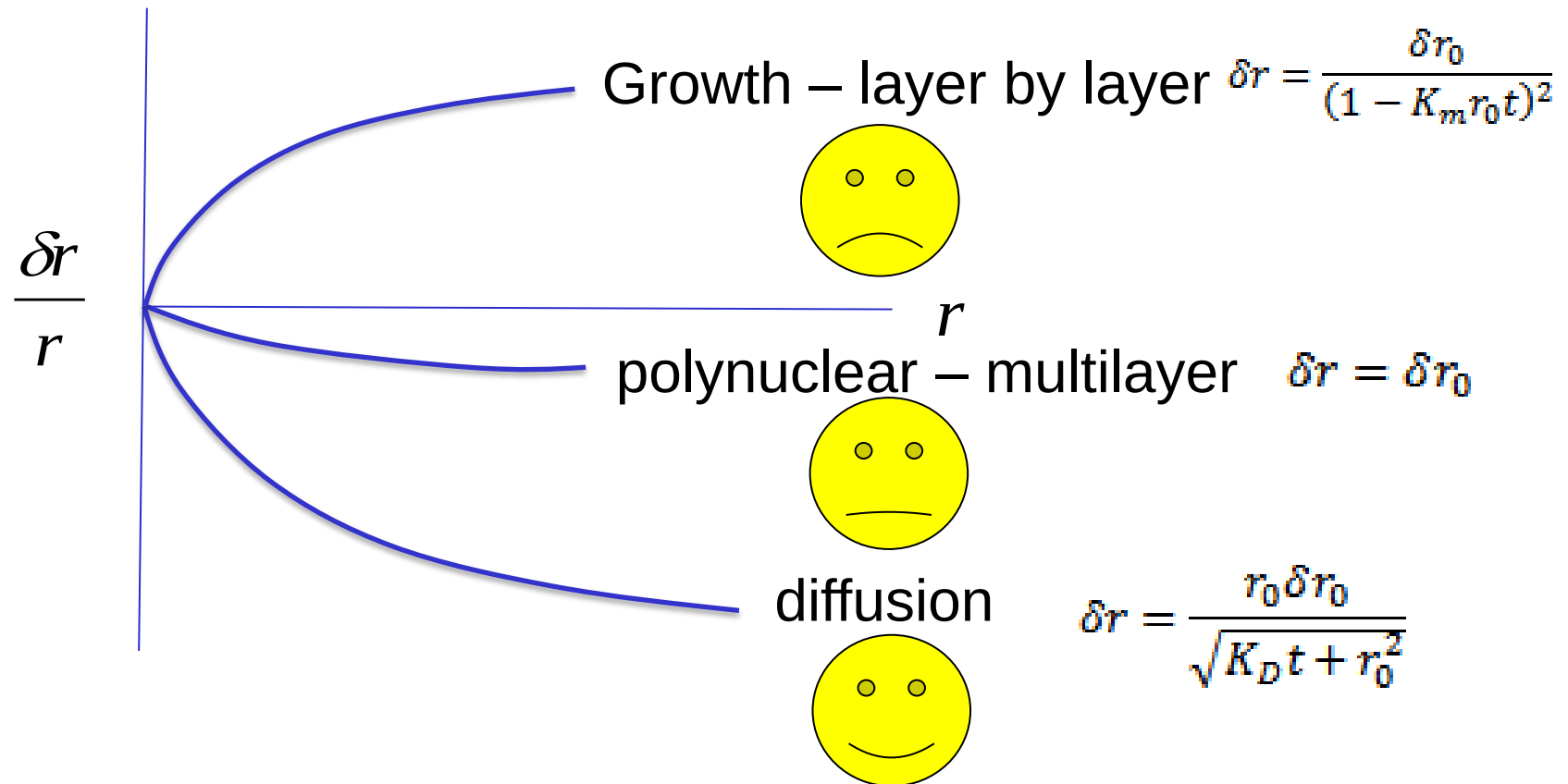
Growth Mechanism of Nuclei

Diffusion-limited and *growth-limited* processes lead to *different size distributions*

To understand these differences, we can computationally evaluate the growth of two particles with different initial nuclear radius (equivalent to particles nucleated at different times)

- Initial radius difference: δr_0
- Radius difference at time, t : δr

Growth Mechanism of Nuclei



Diffusion-limited growth → monosized particles.

Growth Mechanism of Nuclei

How to achieve diffusion-limited growth:

- Extremely low concentration of growth species
- Increased viscosity of solution
- Introduction of a diffusion barrier (e.g., monolayer on surface)
- In reactive processes, control rate of reaction (e.g., concentration, reagent, catalyst)

Controlled supply/delivery makes it possible to manipulate the growth process.

Nanoparticle Size Analysis

Transmission electron microscopy (TEM)

- Direct
- High resolution ~ 0.1 nm (size and shape)
- Technically demanding sample preparation and high cost

Scanning electron microscopy (SEM)

- Direct
- Resolution limited to >1 nm (size and shape)
- Requires conducting samples

Scanning probe microscopy (SPM), e.g., AFM, STM

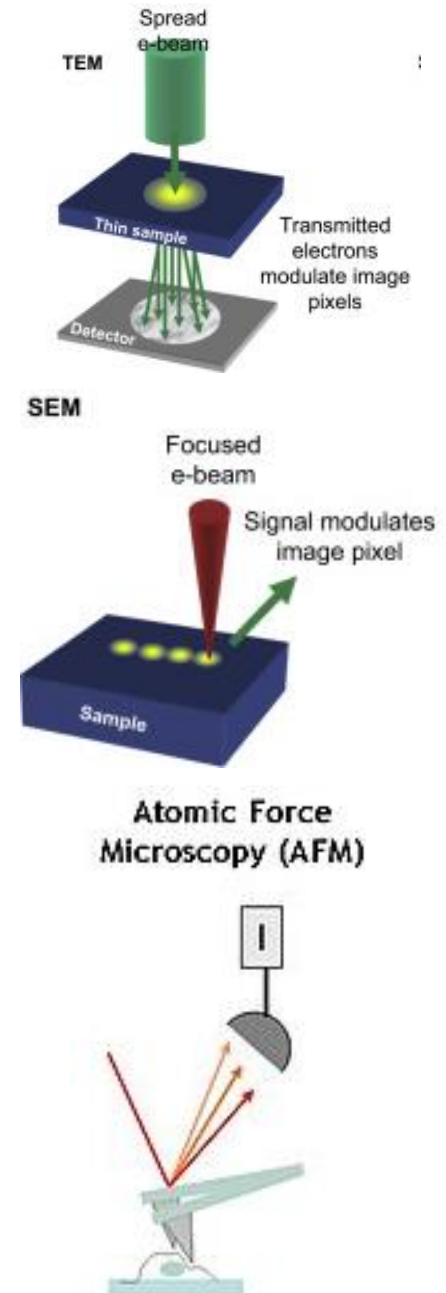
- Direct
- High resolution in z (<0.1 nm), low resolution in x - y
- Tip effects, and often sample-independent

X-ray diffraction (XRD)

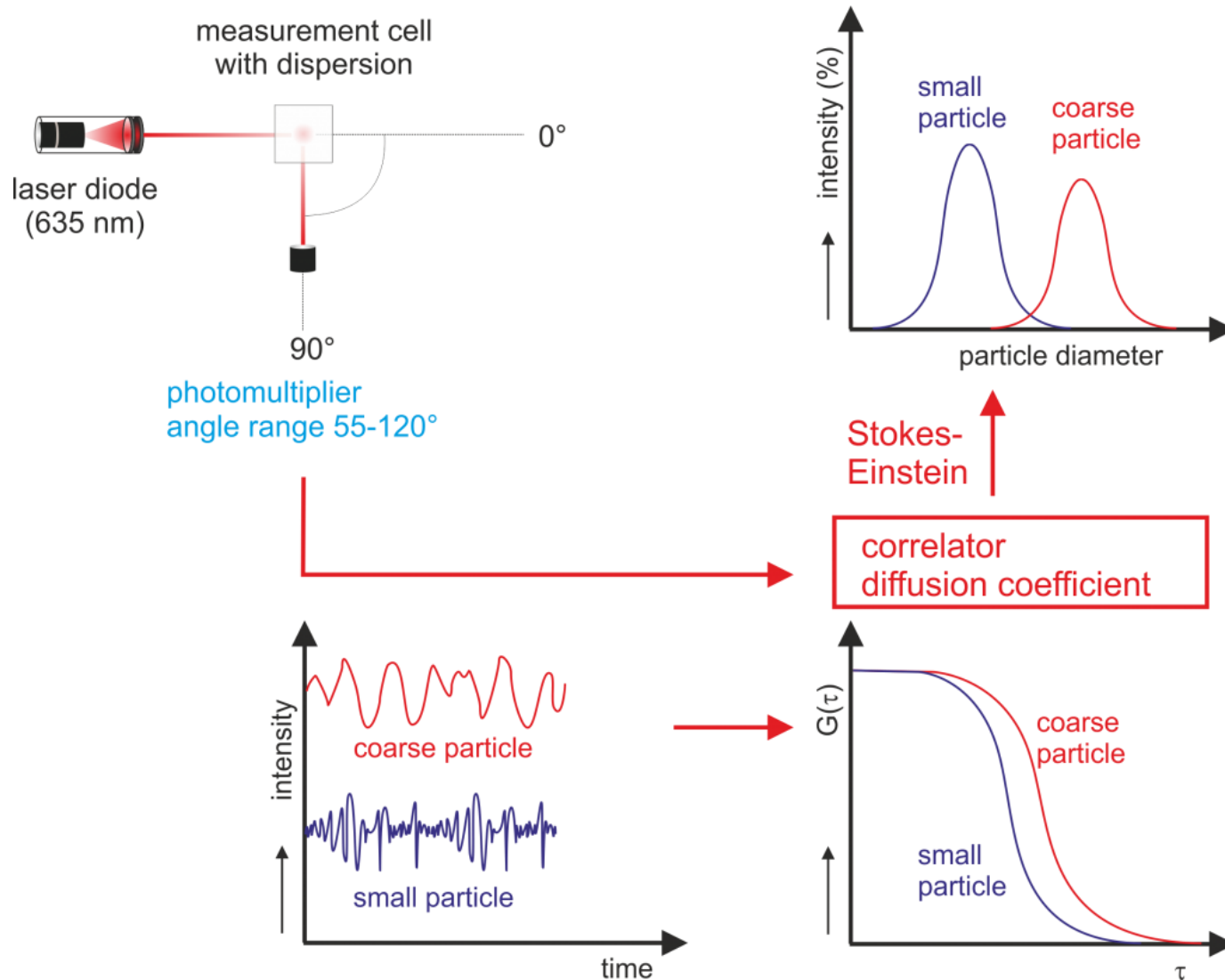
- Indirect
- Not suitable for polycrystalline, core-shell, or non-crystalline NPs, or anisotropic samples

Dynamic light scattering (DLS)

- Indirect
- Medium resolution (3 nm) (size only, no shape)



Dynamic Light Scattering



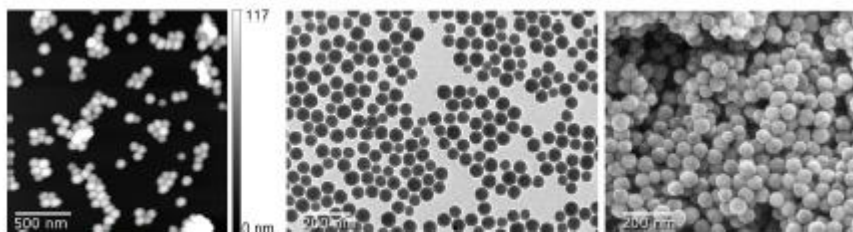


Fig. 1. Examples of silica (SiO₂) nanoparticle images from AFM (left), TEM (centre), and SEM (right).

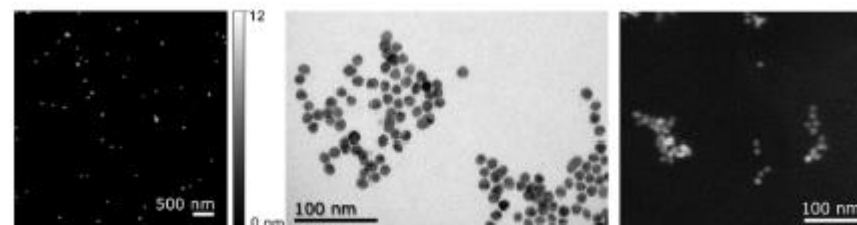
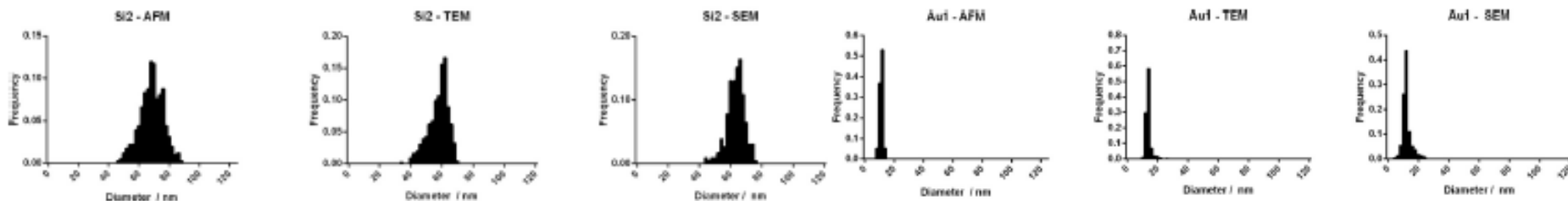


Fig. 2. Examples of gold (Au) nanoparticle images from AFM (left), TEM (centre), and SEM (right).



Technique	Resolution / detection limit	Physical basis	Environment	Material sensitivity	Parameters measured
TEM	0.1nm	Scattering of electrons	Vacuum	Increases with atomic number	Size and shape
SEM	1 nm	Emission of electrons	Vacuum	Somewhat increases with atomic number	Size and shape
AFM	1nm (XY), 0.1 nm (Z)	Physical interaction with sample	Vacuum / air / liquid	Equal for all materials	Size and shape
DLS	3 nm	Light scattering fluctuations due to diffusion	Liquid	Depends on refractive index	Size only

TEM (and AFM) works best for <10 nm NPs

X-ray Diffraction

The finite size of crystallites → peak broadening

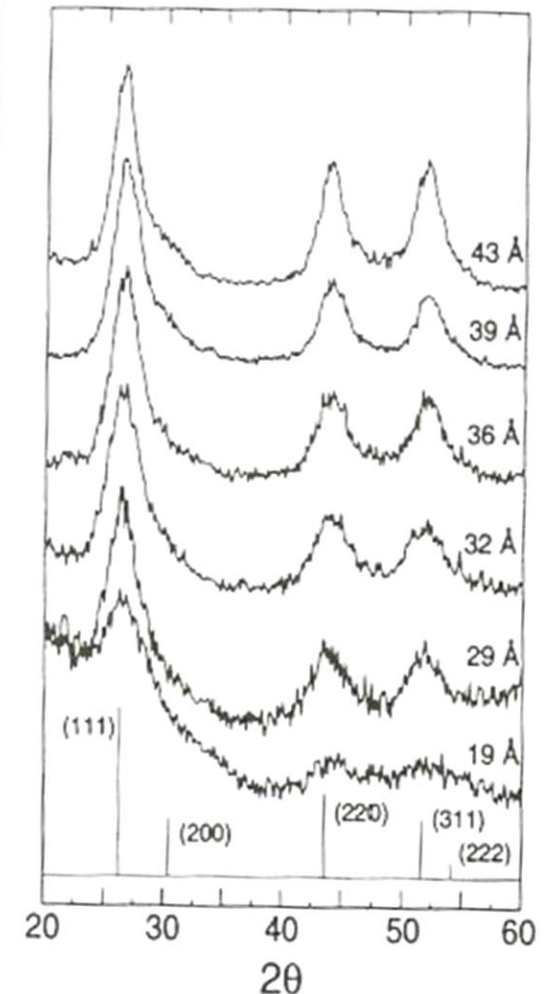
The peak width at half maximum (B) can be used to determine size of crystallites (D) according to Scherrer's formula

$$D = \frac{K\lambda}{B \cos \theta_B}$$

K: Scherrer's constant (typically K~1).

For nanocrystals, the Scherrer formula provides accurate crystallite size information

We will learn about x-ray diffraction in a future topic



Powder XRD of InP nanocrystals with increasing size [J. Phys. Chem. 100, 7212 (1996)]
Stick spectrum gives bulk reflections with relative intensities

X-ray Diffraction vs SEM vs DLS

Table 3.2. Comparison of average sizes of Au nanoparticles synthesized using various reduction reagents, all in nanometer.²⁷

Reduction reagents	436 nm*	546 nm*	XRD [#]	SEM
Sodium citrate	29.1	28.6	17.5	17.6 ± 0.6
Hydrogen peroxide	25.3	23.1	15.1	15.7 ± 1.1
	31.0	31.3	18.7	19.7 ± 2.6
Hydroxylamine hydrochloride			37.8	22.8 ± 4.2
Citric acid	23.5	22.8		12.5 ± 0.6
Carbon monoxide	9.1	7.4	9.0	5.0 ± 0.5
	15.3	15.3	9.8	7.5 ± 0.4
	18.9	18.3	13.1	12.2 ± 0.5
Phosphorus			13.9	8.1 ± 0.5
			21.0	15.5 ± 1.7
			29.6	25.6 ± 2.6
			36.9	35.8 ± 9.7

Further reading on characterization & properties of nanomaterials:

Cao & Wang, **Chapter 8.**

Synthesis of Metallic Nanoparticles

Cao & Wang, Chapter 3, section **3.2.3**.

Synthesis of Semiconductor Nanoparticles

Cao & Wang, Chapter 3, section **3.2.4**.

Synthesis of Oxide Nanoparticles

Cao & Wang, Chapter 3, section **3.2.5**.

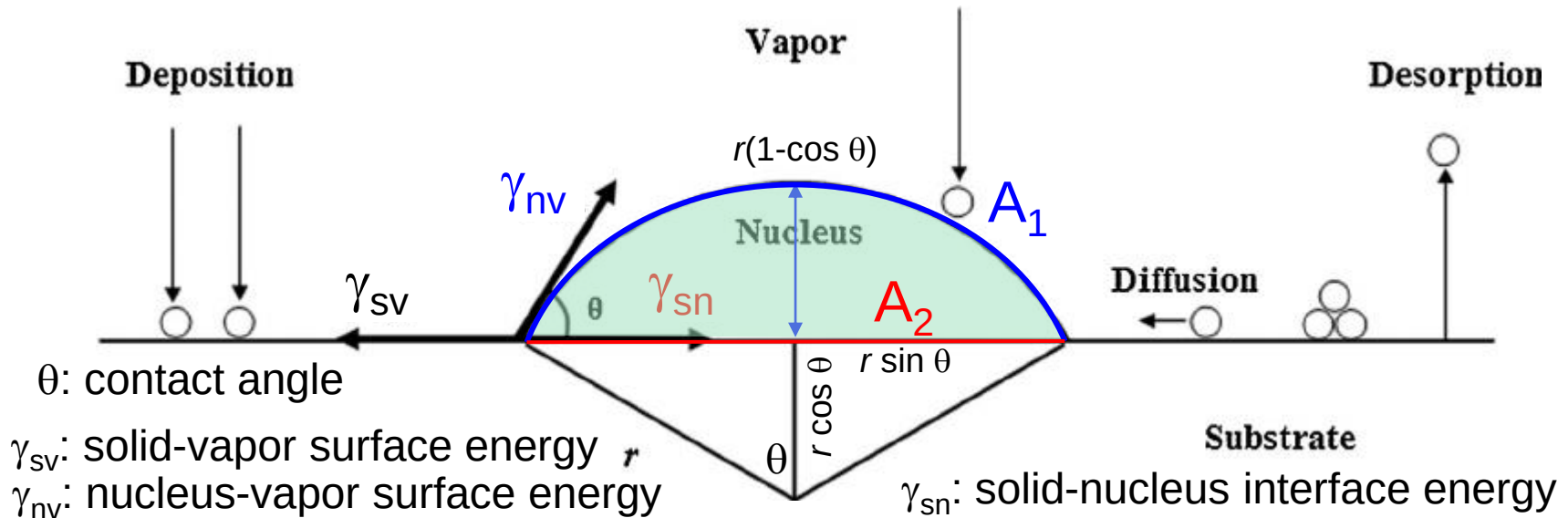
Note: Final student project will be an opportunity to learn more about synthetic details, methods, applications.

Heterogeneous Nucleation

When a new phase forms on the surface of another material, the process is called heterogeneous nucleation

Let's consider a heterogeneous nucleation process on a planar solid surface

Assuming the growth species in solution or vapor impinge the surface, they can diffuse and aggregate to form a nucleus with a cap shape



There is a Gibbs free energy reduction due to the **volume**, but also a gain due to the **surface** and the **interface**

Heterogeneous Nucleation

The total change of Gibbs free energy, ΔG , associated with the formation of this nucleus is

$$\Delta G = a_3 r^3 \Delta G_v + a_1 r^2 \gamma_{nv} + a_2 r^2 \gamma_{sn} - a_2 r^2 \gamma_{sv}$$

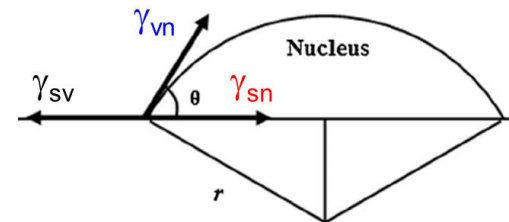
$$a_1 = 2\pi(1 - \cos \theta)$$

$$a_2 = \pi \sin^2 \theta$$

$$a_3 = 3\pi(2 - 3 \cos \theta + \cos^3 \theta)$$

The contact angle, θ , is only dependent on the surface properties of the solid and nucleus, and defined by the *Laplace* equation:

$$\gamma_{sv} = \gamma_{ns} + \gamma_{vn} \cos \theta$$



Heterogeneous Nucleation

$$r^* = -2 \frac{a_1 \gamma_{vn} + a_2 \gamma_{ns} - a_2 \gamma_{sv}}{3a_3 \Delta G_v}$$

$$\Delta G^* = \frac{4(a_1 \gamma_{vn} + a_2 \gamma_{ns} - a_2 \gamma_{sv})^3}{27a_3^2 \Delta G_v^2}$$

Recall: homogeneous nuc.

$$r^* = -2 \frac{\gamma}{\Delta G_v}$$

$$\Delta G^* = \frac{16\pi}{3} \frac{\gamma^3}{\Delta G_v^2}$$

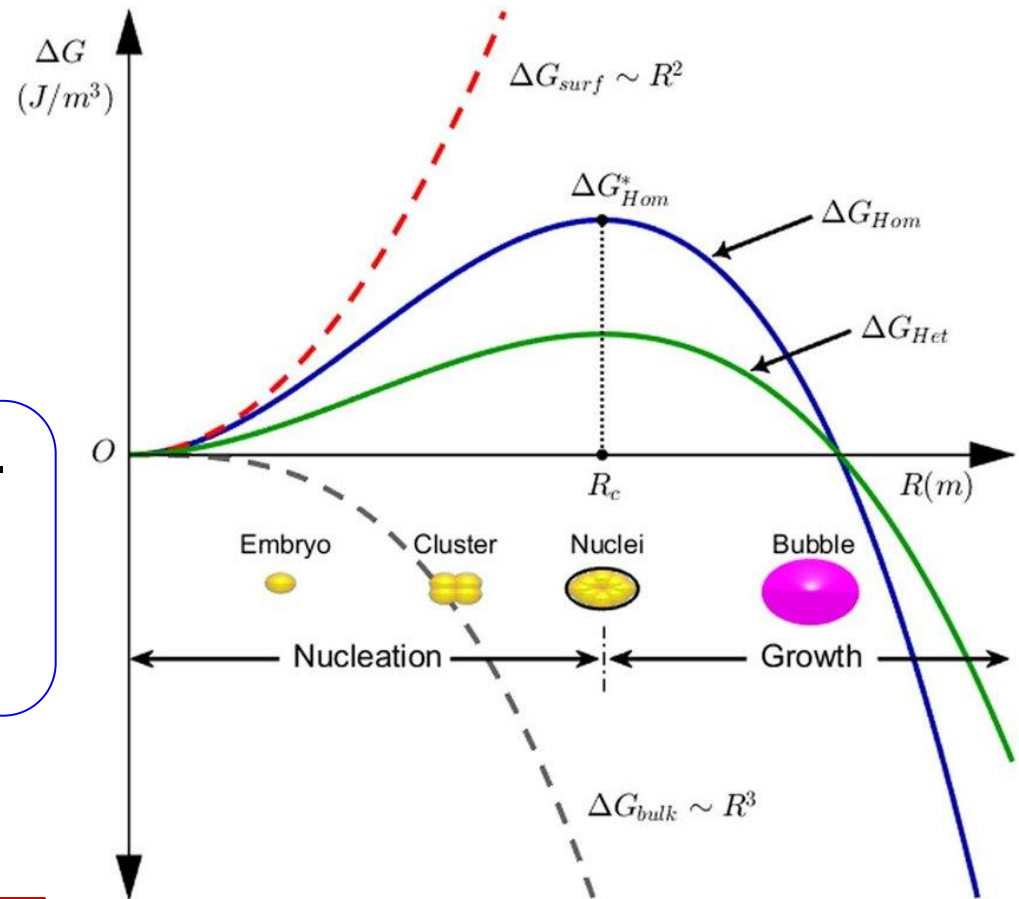
$$r^* = \frac{2\pi\gamma_{vn}}{\Delta G_v} \left(\frac{\sin^2 \theta \cos \theta + 2 \cos \theta - 2}{2 - 3 \cos \theta + \cos^3 \theta} \right)$$

$$\Delta G^* = \frac{16\pi}{3} \frac{\gamma_{vn}^3}{\Delta G_v^2} \left(\frac{2 - 3 \cos \theta + \cos^3 \theta}{4} \right)$$

Wetting factor (≤ 1) $\rightarrow$ **Unity** if $\theta=180^\circ$

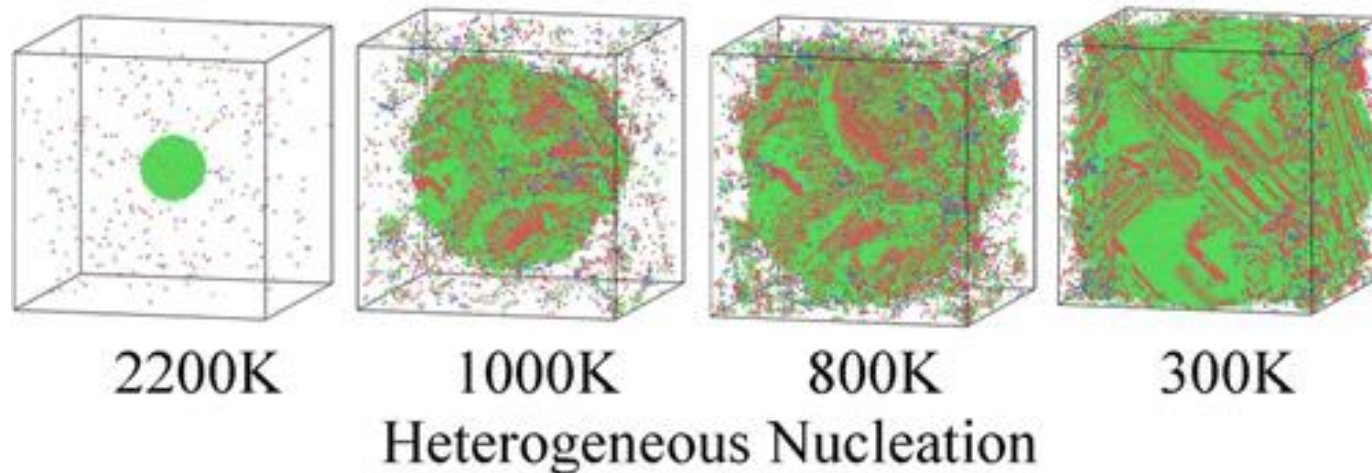
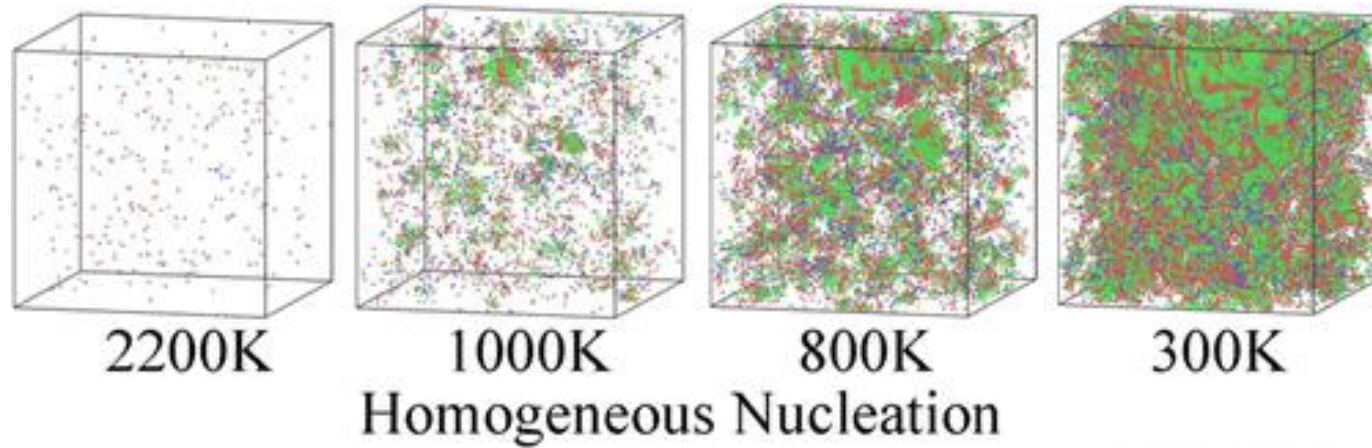
Zero when $\theta=0$ (**no barrier!**)

e.g., same material



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Heterogeneous nucleation always more favorable than homogeneous nucleation.

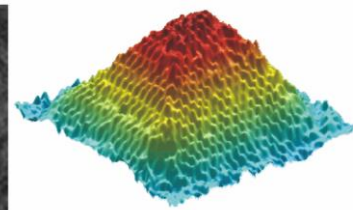
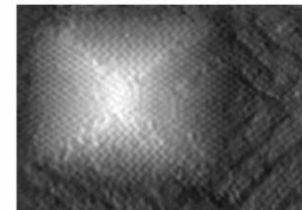
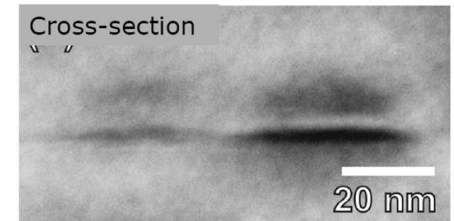
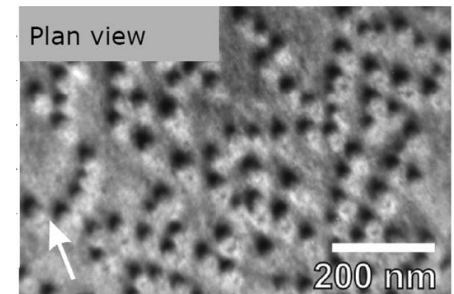
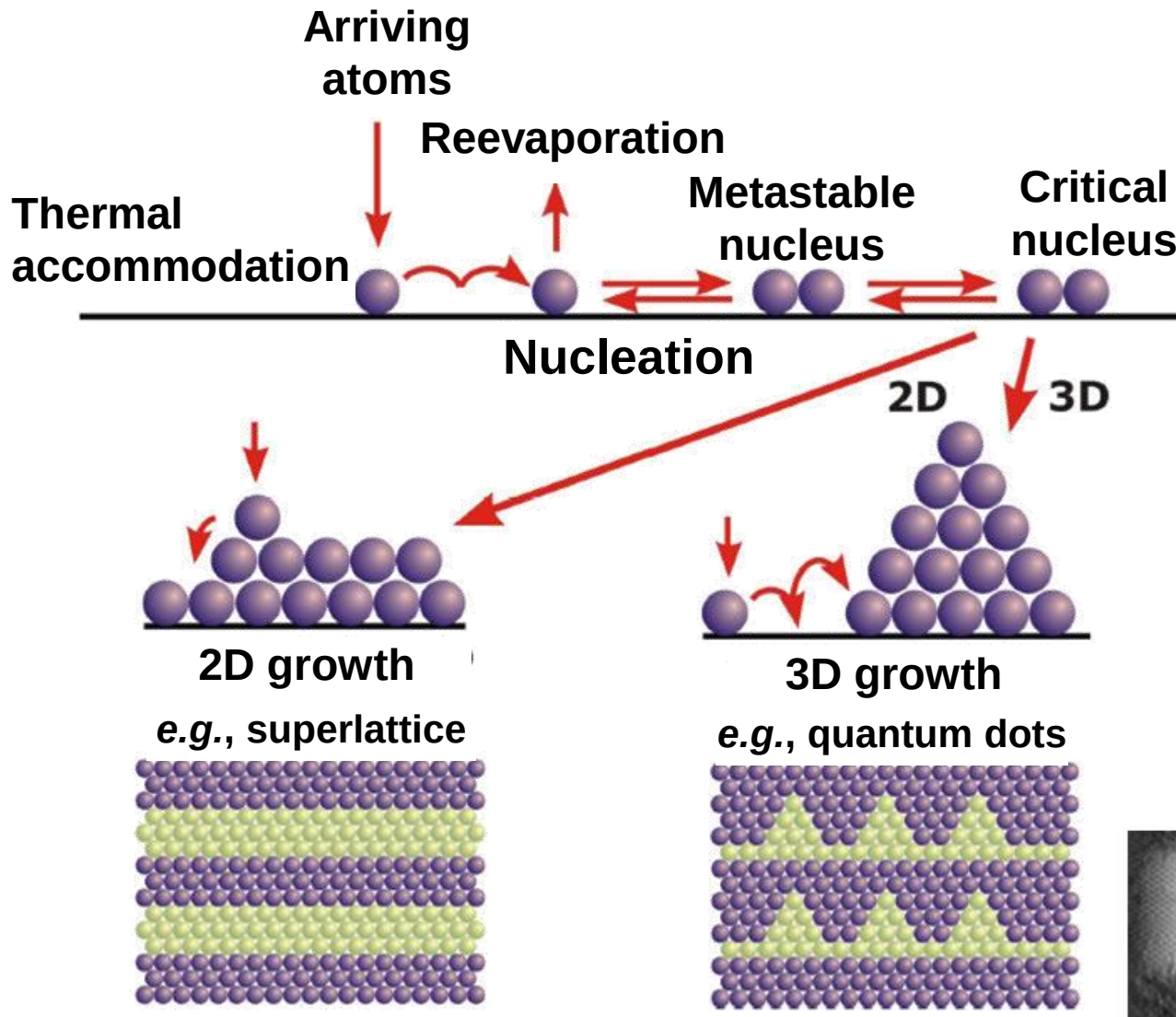


Molecular dynamic simulations

Growth Pattern of Homogeneous and Heterogeneous Nucleation in High-Entropy FeNiCrCoCu Alloys

Cryst. Growth Des. 2022, 22, 4, 2417–2425

Heterogeneous Nucleation



Source: P. Desjardins